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Next item →

1. Which one of the following statements is true? 1 / 1 point

- ☐ A predicate becomes a proposition when its variables are unknown
- ☒ Predicates are a more general form of propositions
- ☐ Predicates behave as functions whose values are always False
- ☐ Predicates behave as functions whose values are always True

✔ Nice work

Correct. As seen in the video, Predicates are used to build more general and complex forms of propositions.

2. Given the statement $S(x)$, where ' x is the capital of France', which one of the following statements is true? 1 / 1 point

- ☒ The property of the statement 'is the capital of France' is called the predicate
- ☐ The subject of the statement x is called the predicate
- ☐ S is said to be the predicate variable
- ☐ When a value is assigned to the variable x , the statement $S(x)$ cannot be described using propositional logic

✔ Nice work

As seen in the video, S is the predicate.

3. Given the statement $S(x)$, where ' x is the capital of France', what is the truth value of $S(\text{Paris})$? 1 / 1 point

- ☒ True
- ☐ False
- ☐ Undefined

✔ Nice work

As Paris is the capital of France, $S(\text{Paris})$ is True

4. Which one of the following statements is true? 1 / 1 point

- ☐ In predicate logic, the $\neg$ operator has a lower precedence than $\wedge$
- ☐ In predicate logic, the $\vee$ operator has lower precedence than $\rightarrow$ and $\leftrightarrow$
- ☐ Logical operations from propositional logic don't apply to predicate logic
- ☒ In predicate logic, the $\wedge$ operator has higher precedence than $\vee$
- ☐ Predicate logic cannot depend on more than one variable

✔ Nice work

As seen in the video, the logical operations discussed in propositional logic apply also to predicate logic. Thus, as for propositional logic, in predicate logic, the $\wedge$ operator has higher precedence than $\vee$.

5. Given the statement $M(x, y, z)$ ' $\frac{x}{y} > z$ ', what is the truth value of $M(10, 5, 2)$? 1 / 1 point

- ☒ False
- ☐ True
- ☐ Undefined

✔ Nice work

$= \frac{10}{5} = 2 \not> 2, M(10, 5, 2)$ is False.

6. Given the statement $P(x)$ ' $\frac{x}{10} > 1$ ', what is the truth value of $P(1) \wedge P(11)$? 1 / 1 point

- ☐ Undefined
- ☒ False
- ☐ True

✔ Nice work

$P(1) = F$ and $P(11) = T$. Hence, $P(1) \wedge P(11) = F \wedge T = F$. This is then False.